

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458875.74664 +/- 0.00069 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.097 +/- 0.044
Transit depth (Rp/Rs)^2	0.94 +/- 0.86 [%]
Orbital Inclination (inc)	86.79 +/- 2.86
Airmass coefficient 1 (a1)	1.17 +/- 0.72
Airmass coefficient 2 (a2)	-0.12 +/- 0.98
Scatter in the residuals of the lightcurve fit is	63.47 %
Best Comparison Star	#3 - [277, 286]
Optimal Aperture	3.0
Optimal Annulus	3.0
Transit Duration (day)	0.1254 +/- 0.0063

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.097 ± 0.044 vs 0.1178 (deviation 0.28σ)
Duration fit vs published	0.1254 d vs 0.125 d (deviation 0.04σ)
Mid-time O-C	-0.2 min
Depth SNR	1.3
Residual scatter	63.47 %

Light curve

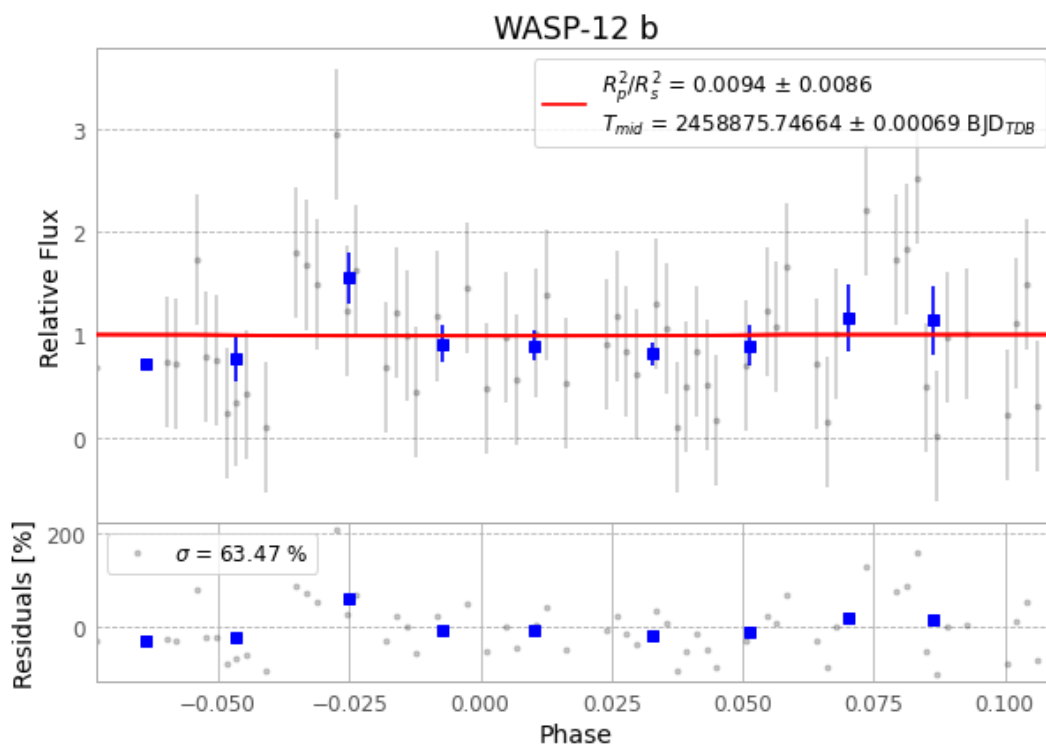


Figure 1 — FinalLightCurve_WASP-12 b_2020-01-26.png (SHA-256 9d92ea3c72ec48e8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [335, 344], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e1f92e43ba9e0040...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

97 FITS frames (64 MB), WASP-12200127035112.FITS → WASP-12200127083912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-200127.log (a42b36ed5e25...), FinalLightCurve_WASP-12 b_2020-01-26.png (9d92ea3c72ec...), FinalLightCurve_WASP-12 b_2020-01-26.csv (88edcc77857c...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 09:39Z.